EE3-23 MACHINE LEARNING EXAM PAPER 2019

Formulas that may be needed:

$$\mathbb{P}\left(|\widehat{R}_n(h) - R(h)| > \varepsilon\right) \leq 4(2n+1)^{d_{VC}} e^{-\varepsilon^2 n/8}$$

1. Practical ML scenario. Consider a fitness tracker for an accurate training program. Design an ML predictor for energy estimation within 10s intervals. Available measurements include the number of steps, the altitude difference, the average heart rate, and the body weight. The processing hardware constrains the predictor to have 9 parameters only.

   Formulate that as an ML problem and design a loss function. Note that exceeding a predefined energy threshold during training due to underestimating of the energy will require 5 times longer recovery time compared to overestimating.

   The ML solution will be useful if it can guarantee that the test error does not differ from the training error by more than 20% with 99% confidence.

   a) Identify relevant ML components and formulate it as an ML problem. [40 marks]

   b) Your colleague advised to use a polynomial transformation of degree 9. Discuss the implications of using this approach. [10 marks]

   c) Propose a hypothesis class for this problem. Calculate the approximate number of data points that are needed for training to make the ML predictions useful. Show your calculations. [25 marks]

   d) What type of regularisation would you use in this problem and why? What other types do you know and why these would be less suitable? Include Lagrangian formulations of the loss for each regularisation you discuss. [25 marks]

2. Consider four data points $\mathbf{x}_1 = (0,-4)$, $\mathbf{x}_2 = (2,0)$, $\mathbf{x}_3 = (0,4)$, $\mathbf{x}_4 = (-2,0)$ and their labels $y_1 = -1$, $y_2 = -1$, $y_3 = 1$,$y_4 = 1$.

a) Derive the solution formula for $\mathbf{w}$ in a linear regression problem using matrix formulation. [20 marks]

b) Input the data points $\mathbf{x}_{1...,4}$ with their labels $y_{1...,4}$ in the formula resulting from question (a) and calculate the values of $\mathbf{w}$ for this regression problem. [15 marks]

c) When can the method for solving linear regression be used to find a solution for a binary classification? Give an example and discuss the benefits and risks. [15 marks]

d) Using the four data points $\mathbf{x}_{1...,4}$ with their labels $y_{1...,4}$ write down the solution for a ridge regression with $\lambda = 0.1$, and calculate values $\mathbf{w}_r$. [15 marks]

e) Calculate the predictions of a linear regressor with parameters $\mathbf{w} = (0,-1,0.25)$ for the four data points $\mathbf{x}_{1...,4}$ and the error of this predictor. [10 marks]

f) Logistic regression function. Write down its formula, its error function, and draw the curve. [10 marks]

g) Design a Multi-Layer-Perceptron (MLP) architecture with 1 hidden layer (include the weights) to model the logical operation of $b_1\bar{b}_2 + \bar{b}_1 b_2$ with $b_i \in \{-1,1\}$ and perceptron output $y_i \in \{-1,1\}$. [15 marks]

3. Support Vector Machines

Consider that we have 6 data samples with a single feature, where the samples $\{-2,-1,0\}$ have label -1, while the samples $\{-4,3,4\}$ have label $+1$.

a) Plot the dataset, and explain if it can be perfectly separated using a linear separator (in the available feature space), or not. [10 marks]

b) We define a feature map $z = \Phi(x) = (x, x^2)$, which maps each data sample to a two-dimensional feature vector. Apply this mapping to the dataset, and plot the corresponding points in the new feature space. [20 marks]

c) Write down the kernel that corresponds to the feature mapping $\Phi(x)$ given above. [20 marks]

d) Construct the separating hyperplane with the maximum margin. Remember that this hyperplane can be expressed as $\mathbf{w}^T\mathbf{x} + b = 0$ with $\|\mathbf{w}\| = 1$, where $\mathbf{x}$ refers to a two-dimensional point in the new feature space. Find the values for $\mathbf{w}$ and b. What is the corresponding margin, and which are the support vectors? (Hint: Maximum margin hyperplane in general can be found solving a quadratic program, but in this particular case you can identify the solution by purely geometric arguments.) [40 marks]

e) If you are given two new data samples with values 5 and -5, both labeled as $+1$, what would be the new values of `w` and `b`? [10 marks]

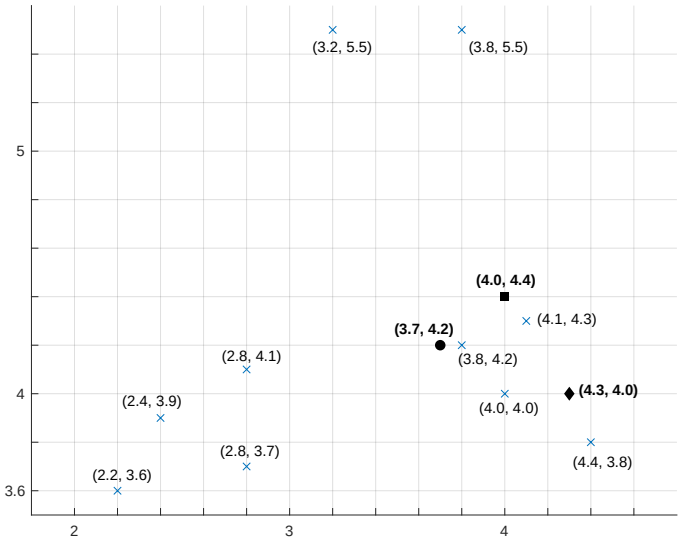
4. a) Clustering

We are given 10 data samples, shown in the figure above with crosses ('x'):

$$\{(2.2,3.6),(2.4,3.9),(2.8,3.7),(2.8,4.1),(3.2,5.5),$$

$$(3.8,4.2),(3.8,5.5),(4.0,4.0),(4.1,4.3),(4.4,3.8)\}$$

We would like to cluster these data samples into $k = 3$ clusters using k-means clustering, using the Euclidean distance as the distance function. The initial cluster centers are given by $(3.7,4.2)$ shown by • in the figure, $(4.0,4.4)$ shown by ■, and $(4.3,4.0)$ shown by ♦.



i) What is the center of the cluster '•' after one iteration? [10 marks]

ii) What is the center of the cluster '■' after two iterations? [10 marks]

iii) How many iterations are needed for convergence? List the points in all three clusters after convergence. [15 marks]

b) Reinforcement Learning

We will consider the following simple model for flying a drone. The drone can fly slowly or fast. If it flies fast, it receives an immediate reward of 2, and if it flies slowly, it receives a reward of 1. The battery of the drone has three states, high (H), low (L), and empty (E). If the battery is empty, the drone cannot operate any more; hence, it is a terminal state with zero reward. The drone harvests solar energy so with some probability it gets charged as it flies. If the battery is low, and it flies fast, with probability $1/9$ the battery becomes empty, and with probability $8/9$ it remains low. If the battery is low, and it flies slowly, with probability $1/3$ the battery becomes high, and with probability $2/3$ it remains low. If the battery is high, and it flies slowly, the battery remains high. If the battery is high, and it flies fast, with probability $1/3$ the battery remains high, and with probability $2/3$ it becomes low.

i) Is this a Markov decision process? Explain why. How many states are there? Draw the state transition diagram, indicating the actions, transition probabilities and rewards. [15 marks]

ii) Consider the policy that the drone always flies slowly. Write down the Bellman equation for the state-value function, and evaluate the values of all the states, for a discount rate of $\gamma = 0.9$. [20 marks]

iii) Consider the policy that the drone flies fast when battery is high, and flies slowly when the battery is low. Find the state value function $v_1(s)$ for this policy. Then apply one iteration of the greedy policy improvement to compute a new deterministic policy $v_2(s)$ (with discount factor $\gamma = 0.9$). Is the value function $v_2(s)$ optimal? Explain your answer. [30 marks]